

Cloud LearnX

Mathematics

Transformations and Bearings — Practice Worksheet

Student: Amy

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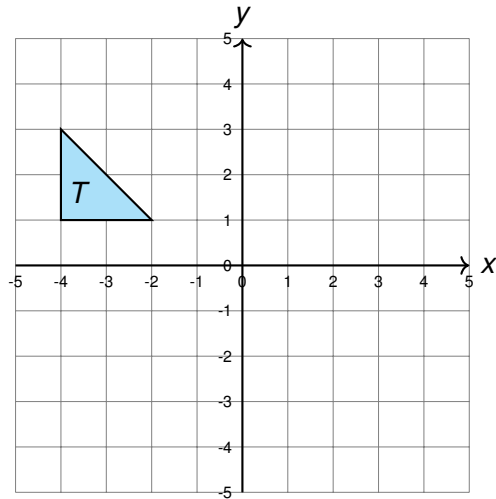
Topic: Transformations and Bearings

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.
Write your answers in the spaces provided.
You must write down all the stages in your working.

1 Triangle T is drawn on the grid below.



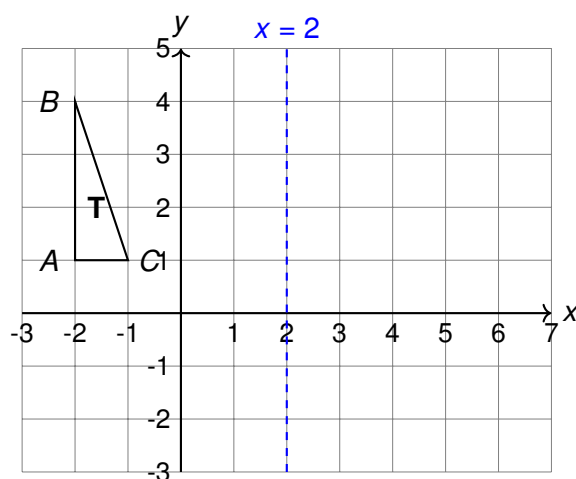
Translate triangle T by the vector $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$.

Draw the image of triangle T on the grid and label it T' .

(2)

(Total for Question 1 is 2 marks)

- 2 A UI/UX designer is prototyping a symmetrical icon on a grid-based design canvas. She has drawn triangle T and wants to check the mirror-image icon by reflecting it in the vertical line $x = 2$.



Triangle T has vertices $A(-2, 1)$, $B(-2, 4)$ and $C(-1, 1)$.

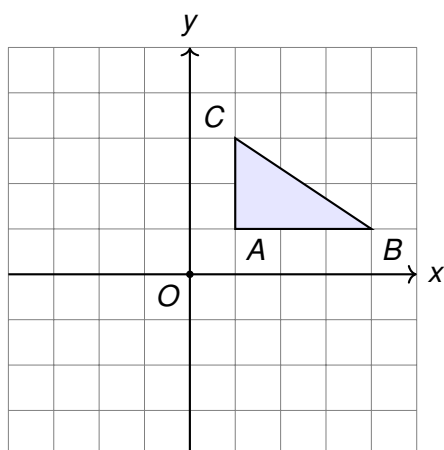
Reflect triangle T in the line $x = 2$.

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(2)

(Total for Question 2 is 2 marks)

3 Triangle S has vertices at $A(1, 1)$, $B(4, 1)$ and $C(1, 3)$.

Triangle S is rotated 90° clockwise about the origin O to give triangle S' .



On the grid, draw triangle S' , the image of triangle S after a rotation of 90° clockwise about the origin O .

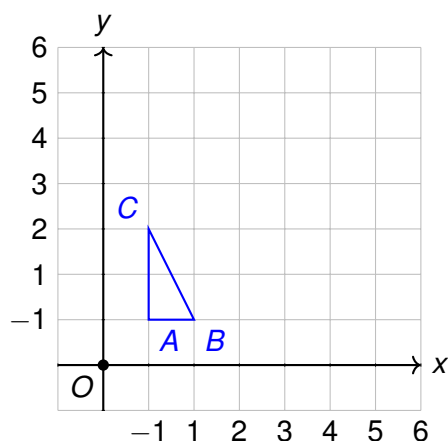
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(3)

(Total for Question 3 is 3 marks)

4 Triangle P has vertices $A(1, 1)$, $B(2, 1)$ and $C(1, 3)$.

Triangle P is enlarged by scale factor 2, using centre of enlargement $(0, 0)$, to give triangle P' .

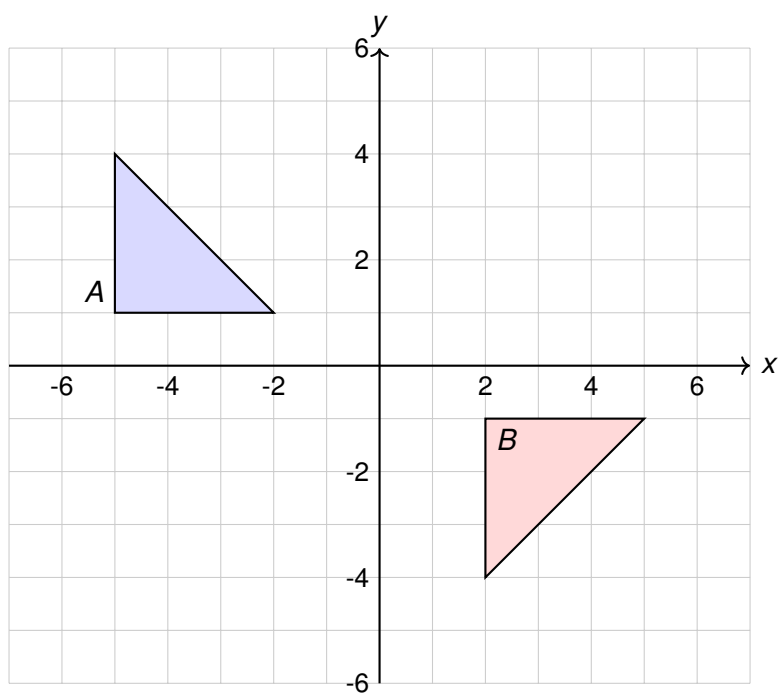


On a copy of this grid, draw and label triangle P' , the image of triangle P after the enlargement.

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(3)

(Total for Question 4 is 3 marks)

- 5 Triangle A has vertices at $(-5, 1)$, $(-5, 4)$ and $(-2, 1)$. Triangle B has vertices at $(2, -1)$, $(2, -4)$ and $(5, -1)$.

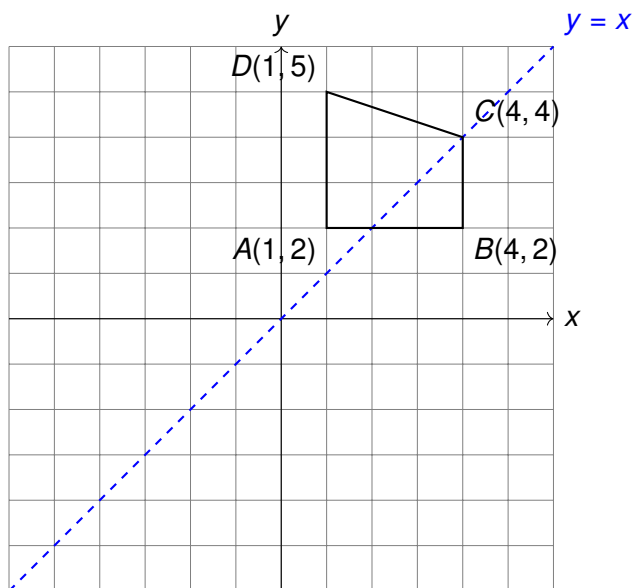


Describe fully the single transformation that maps triangle A onto triangle B .

(3)

(Total for Question 5 is 3 marks)

- 6 A UI/UX designer is creating a symmetrical icon for an app. She draws quadrilateral Q on a coordinate grid, then reflects it in the line $y = x$ to create a mirror-image icon Q' .



Quadrilateral Q has vertices $A(1, 2)$, $B(4, 2)$, $C(4, 4)$ and $D(1, 5)$.

Reflect quadrilateral Q in the line $y = x$ to give quadrilateral Q' .

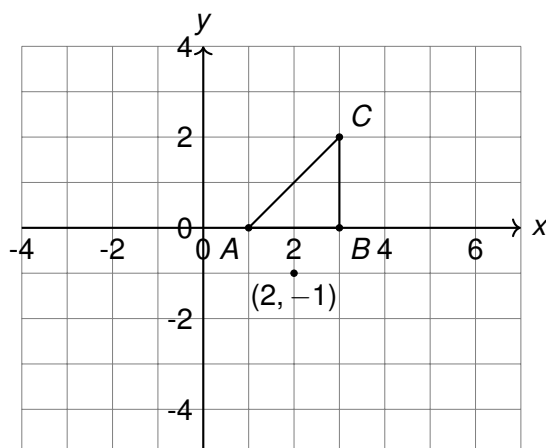
Write down the coordinates of the vertices of Q' .

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(2)

(Total for Question 6 is 2 marks)

7 Triangle L has vertices $A(1, 0)$, $B(3, 0)$ and $C(3, 2)$, as shown on the grid.



Rotate triangle L by 180° about the point $(2, -1)$.

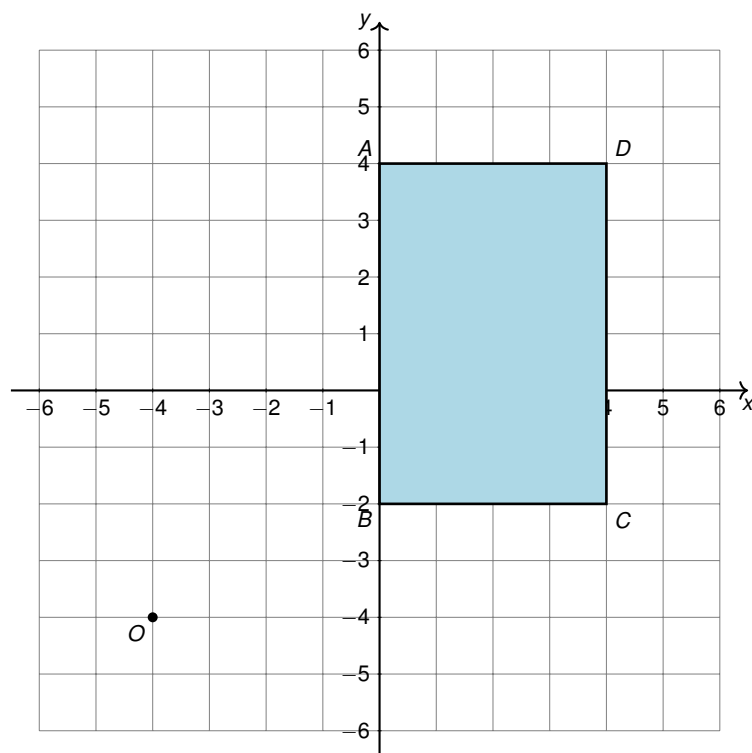
Draw the image of triangle L and label it L' .

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(3)

(Total for Question 7 is 3 marks)

8 Shape Q is shown on the grid below. Point O is marked as a centre of enlargement.



Enlarge shape Q by scale factor $\frac{1}{2}$, centre of enlargement $O(-4, -4)$.

(a) Work out the coordinates of the image of each vertex of shape Q .

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(2)

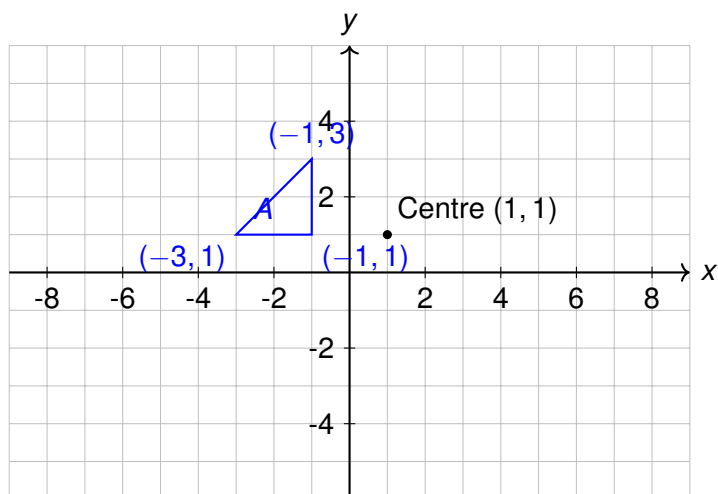
(b) On the grid above, draw the enlarged shape Q' .

(1)

(Total for Question 8 is 3 marks)

9 Triangle A has vertices at $(-1, 1)$, $(-1, 3)$ and $(-3, 1)$.

On the grid below, enlarge triangle A by scale factor -2 , using $(1, 1)$ as the centre of enlargement. Label the image A' .



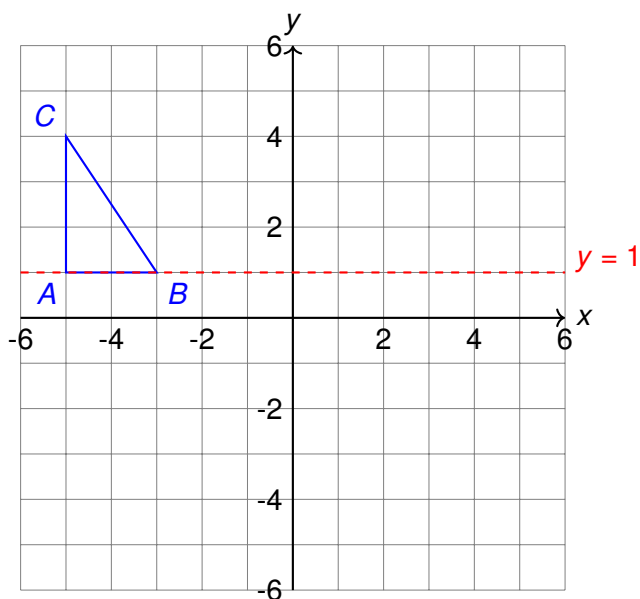
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(4)

(Total for Question 9 is 4 marks)

10 Triangle T has vertices $A(-5, 1)$, $B(-3, 1)$ and $C(-5, 4)$.

Triangle T is first translated by the vector $\begin{pmatrix} 6 \\ -3 \end{pmatrix}$ to give triangle U .

Triangle U is then reflected in the line $y = 1$ to give triangle V .



(a) On the grid, draw and label triangle U , the image of triangle T after the translation.

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(2)

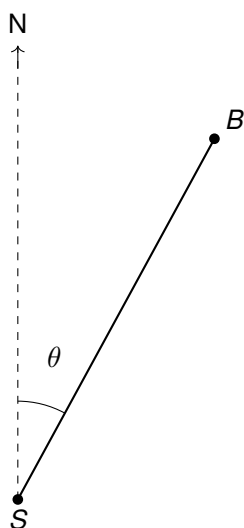
(b) On the grid, draw and label triangle V , the image of triangle U after the reflection.

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(2)

(Total for Question 10 is 4 marks)

11 A weather station, S , is tracking a weather balloon, B , using radar.

The scale diagram below shows the position of the weather station S and the position of the balloon B , using a scale of 1 cm : 5 km. North is shown at S .



Measure the bearing of the balloon B from the weather station S .

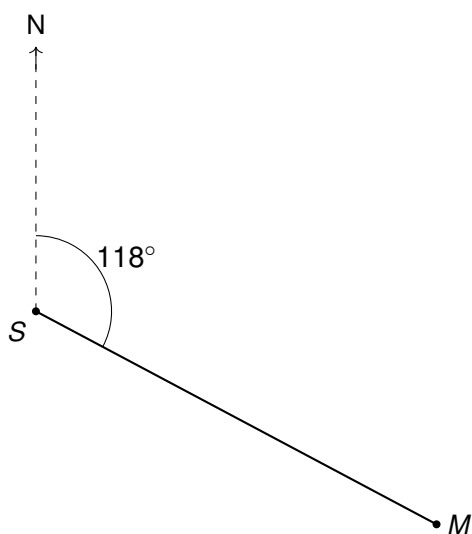
Give your answer to the nearest degree.

$\theta = \dots\dots\dots^\circ$

(3)

(Total for Question 11 is 3 marks)

- 12** A marine science research base, S , sits on the coast. A research ship sails from S on a bearing of 118° to reach a monitoring buoy at point M , where it collects water samples.



Work out the bearing of the marine science base S from the buoy M .

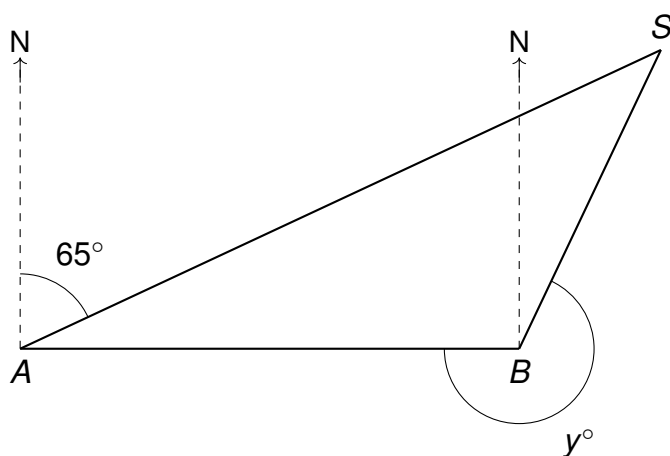
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(3)

(Total for Question 12 is 3 marks)

- 13** Two space observatories, A and B , are tracking a satellite, S , as it passes overhead. Observatory B is due east of Observatory A .

The bearing of the satellite S from Observatory A is 065° .

The bearing of the satellite S from Observatory B is 312° .



- (a) Show that the angle marked y° in the diagram, angle ABS , is 42° .

(2)

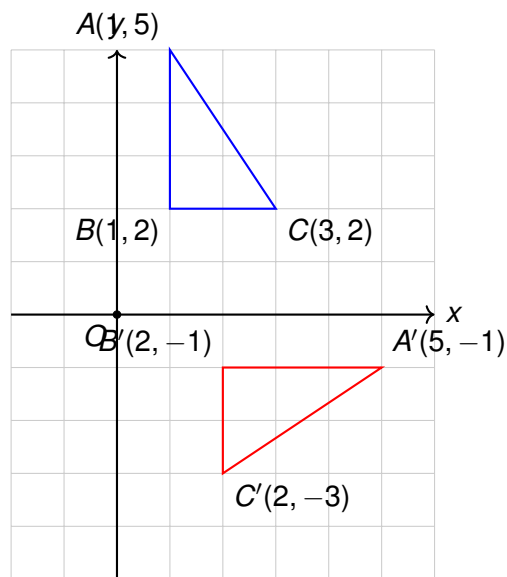
- (b) Hence, using angle facts for the parallel North lines at A and B , work out the bearing of Observatory A from the satellite S .

= $^\circ$

(2)

(Total for Question 13 is 4 marks)

- 14** In a digital art app, a designer creates a triangular icon ABC on a coordinate grid, then rotates it 90° clockwise about the point $(0, 0)$ to produce triangle $A'B'C'$ for a UI animation.



Triangle ABC has vertices $A(1, 5)$, $B(1, 2)$ and $C(3, 2)$.

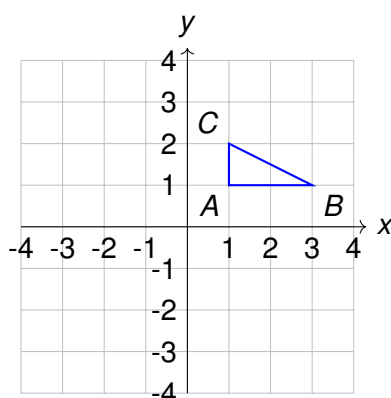
Triangle $A'B'C'$ has vertices $A'(5, -1)$, $B'(2, -1)$ and $C'(2, -3)$.

Show that triangle ABC is congruent to triangle $A'B'C'$ by comparing the lengths of all three corresponding sides.

(3)

(Total for Question 14 is 3 marks)

- 15** A UI/UX designer is prototyping an icon set on a grid-based canvas. Triangle J has vertices $A(1, 1)$, $B(3, 1)$ and $C(1, 2)$, as shown on the grid below.



- (a) Triangle J is enlarged by scale factor -1 , centre $(0, 0)$, to give triangle K .
On a copy of the grid, draw and label triangle K , and state the coordinates of its vertices A' , B' and C' .

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(3)

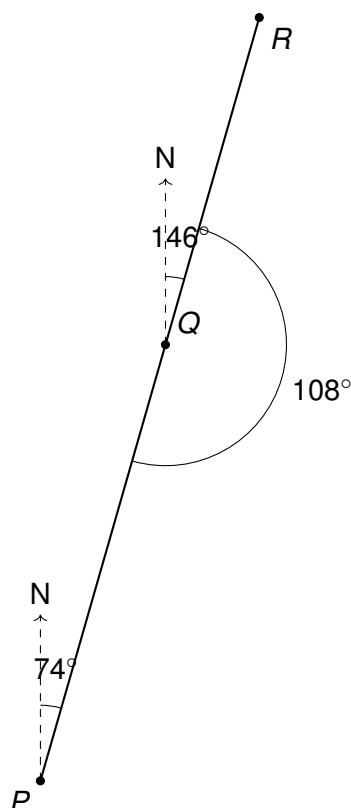
- (b) Describe fully the single transformation that maps triangle J directly onto triangle K .

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(2)

(Total for Question 15 is 5 marks)

- 16** Three monitoring stations, P , Q and R , are set up by a science field team to track wildlife movement across a nature reserve. The stations form a triangle, as shown in the diagram (not drawn accurately).
 The bearing of Q from P is 074° .
 The bearing of R from Q is 146° .
 Angle $PQR = 108^\circ$.



Calculate the bearing of R from P .

$$\theta = \dots\dots\dots^\circ$$

(5)

(Total for Question 16 is 5 marks)

TOTAL FOR PAPER IS ?? MARKS